Name
Date
Course

Absolute Value Equations and Inequalities

Absolute Value

Example I) $|3| = \boxed{3}$

Example II) $|-5| = \boxed{5}$

Example III) $|36| = \boxed{36}$

Example IV) $|-1047| = \boxed{1047}$

Absolute Value Equations

Solve.

① $|x| = 8$

$\boxed{x=8 \text{ or } x=-8}$

② $|y| = 3$

$\boxed{y=3 \text{ or } y=-3}$

③ $|b|+3=7$

$|b| = 4$

$\boxed{b=4 \text{ or } b=-4}$

④ $10 = |x|-2$

$12 = |x|$

$\boxed{12=x \text{ or } -12=x}$

⑤ $\frac{|y|}{4} = 2$

$|y| = 8$

$\boxed{y=8 \text{ or } y=-8}$

⑥ $-2 = \frac{|b|}{-5}$

$10 = |b|$

$\boxed{10=b \text{ or } -10=b}$

⑦ $9 = 2|x|$

$\frac{9}{2} = |x|$

$\boxed{\frac{9}{2}=x \text{ or } -\frac{9}{2}=x}$

⑧ $5|y| = 10$

$|y| = 2$

$\boxed{y=2 \text{ or } y=-2}$

⑨ $3|x|-7=4$

$7+3|x|-7 = 4+7$

$3|x| = 11$

$\frac{3|x|}{3} = \frac{11}{3}$

$|x| = \frac{11}{3}$

$\boxed{x=11/3 \text{ or } x=-\frac{11}{3}}$

⑩ $15 = 3+\frac{|y|}{2}$

$24 = |y|$

$\boxed{24=y \text{ or } -24=y}$

⑪ $7 = 2|b| + 5$

$1 = |b|$

$\boxed{b = 1 \text{ or } b = -1}$

⑫ $\frac{|x|}{3} - 8 = 1$

$|x| = 27$

$\boxed{x = 27 \text{ or } x = -27}$

⑬ $-6 = 1 - \frac{|b|}{3}$

$-6 = 1 + \frac{|b|}{-3}$

$-7 = \frac{|b|}{-3}$

$21 = |b|$

$\boxed{21 = b \text{ or } -21 = b}$

⑭ $0 = 2 - |x|$

$-2 + 0 = 2 - |x| - 2$

$-2 = -|x|$

$\frac{-2}{-1} = \frac{(-1)|x|}{-1}$

$2 = |x|$

$\boxed{x = 2 \text{ or } x = -2}$

⑮ $7 - |b| = 4$

$-|b| = -3$

$(-1)|b| = -3$

$|b| = 3$

$\boxed{b = 3 \text{ or } b = -3}$

⑯ $4 - 2|y| = 3$

$-2|y| = -1$

$|y| = \frac{1}{2}$

$\boxed{y = \frac{1}{2} \text{ or } y = -\frac{1}{2}}$

⑰ $|y| + 2 = -3$

$|y| = -5$

$\boxed{\text{No Solution}}$

⑱ $4 = \frac{|b|}{-3}$

$-12 = |b|$

$\boxed{\text{No Solution}}$

⑲ $|x + 2| = 7$

$x + 2 = 7 \text{ or } x + 2 = -7$

$\boxed{x = 5 \text{ or } x = -9}$

⑳ $4 = |3y|$

$4 = 3y \text{ or } -4 = 3y$

$\boxed{\frac{4}{3} = y \text{ or } -\frac{4}{3} = y}$

㉑ $2 = |-3 + b|$

$2 = -3 + b \text{ or } -2 = -3 + b$

$\boxed{5 = b \text{ or } 1 = b}$

㉒ $\left|\frac{x}{-3}\right| = 10$

$\frac{x}{-3} = 10 \text{ or } \frac{x}{-3} = -10$

$\boxed{x = -30 \text{ or } x = 30}$

(23) $3|5x-2|+4=10$

$-4+3|5x-2|+4=10-4$

$3|5x-2|=6$

$\frac{3|5x-2|}{3}=\frac{6}{3}$

$|5x-2|=2$

$5x-2=2$ or $5x-2=-2$

$\boxed{x=\frac{4}{5} \text{ or } x=0}$

(24) $22=6\left|\frac{y}{5}+3\right|-2$

$4=\left|\frac{y}{5}+3\right|$

$4=\frac{y}{5}+3$ or $-4=\frac{y}{5}+3$

$\boxed{5=y \text{ or } -35=y}$

(25) $\frac{\left|\frac{b}{3}-4\right|}{2}-5=-5$

$5+\frac{|b/3-4|}{2}-5=-5+5$

$\frac{|b/3-4|}{2}=0$

$(2)\frac{|b/3-4|}{2}=0(2)$

$|b/3-4|=0$

$b/3-4=0$

$\boxed{b=12}$

(26) $4|3+2x|-8=4$

$|3+2x|=3$

$3+2x=3$ or $3+2x=-3$

$\boxed{x=0 \text{ or } x=-3}$

(27) $-5=\frac{|8b-1|}{-4}-3$

$8=|8b-1|$

$8=8b-1$ or $-8=8b-1$

$\boxed{\frac{9}{8}=b \text{ or } \frac{-7}{8}=b}$

(28) $8=6+\frac{|2-4y|}{5}$

$10=|2-4y|$

$10=2-4y$ or $-10=2-4y$

$\boxed{-2=y \text{ or } 3=y}$

(29) $-5\left|\frac{x}{6}+7\right|+14=14$

$\left|\frac{x}{6}+7\right|=0$

$\frac{x}{6}+7=0$

$\boxed{x=-42}$

(30) $16=8+\frac{\left|\frac{y}{-2}-3\right|}{5}$

$40=\left|\frac{y}{-2}-3\right|$

$40=\frac{y}{-2}-3$ or $-40=\frac{y}{-2}-3$

$\boxed{-86=y \text{ or } 74=y}$

(31) $-7|1-4x|+8=5$

$|1-4x|=\frac{3}{7}$

$1-4x=\frac{3}{7}$ or $1-4x=-\frac{3}{7}$

$-4x=-\frac{4}{7}$ or $-4x=-\frac{10}{7}$

$\boxed{x=\frac{1}{7} \text{ or } x=\frac{5}{14}}$

(32) $-9\left|\frac{y}{5}+6\right|-8=-12$

$\left|\frac{y}{5}+6\right|=\frac{4}{9}$

$\frac{y}{5}+6=\frac{4}{9}$ or $\frac{y}{5}+6=-\frac{4}{9}$

$\frac{y}{5}=\frac{-50}{9}$ or $\frac{y}{5}=\frac{-58}{9}$

$\boxed{y=\frac{-250}{9} \text{ or } y=\frac{-290}{9}}$

(33) $8=-5+\frac{3|y-1|}{2}$

$5+8=-5+\frac{3|y-1|}{2}+5$

$13=\frac{3|y-1|}{2}$

$(2)13=\frac{3|y-1|}{2}(2)$

$26=3|y-1|$

$\frac{26}{3}=\frac{3|y-1|}{3}$

$26/3=|y-1|$

$y-1=26/3$ or $y-1=-\frac{26}{3}$

$\boxed{y=29/3 \text{ or } y=-23/3}$

(34) $\frac{5|x+2|}{3}+4=6$

$5|x+2|=6$

$|x+2|=6/5$

$x+2=\frac{6}{5}$ or $x+2=-\frac{6}{5}$

$\boxed{x=-4/5 \text{ or } x=-\frac{16}{5}}$

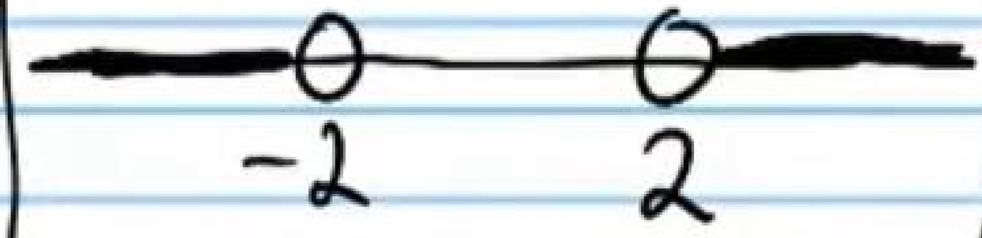
Absolute Value Inequalities

(also called compound inequalities)

Find the algebraic solutions and the graphical solutions of the following inequalities:

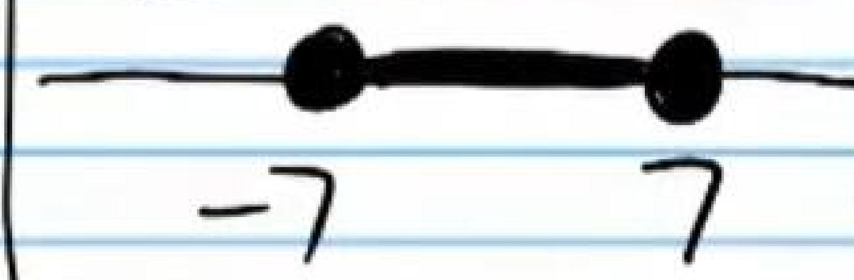
(35) $|x|>2$

$x<-2$ or $x>2$



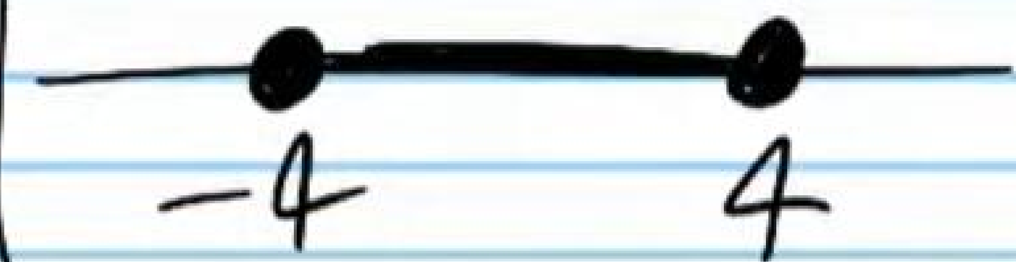
(36) $7\geq|y|$

$y\geq-7$ and $y\leq7$



(37) $|b|\leq4$

$b\geq-4$ and $b\leq4$



(38) $8<|x|$

$x<-8$ or $x>8$

(39) $|y|>-7$

Y can be any real number.

(40) $-3>|x|$

No solution

(41) $|b|>0$

$\boxed{\text{b can be any real number except zero.}}$

(42) $0\geq|y|$

$\boxed{y=0}$

(43) $0>|x|$

$\boxed{\text{No solutions}}$

(44) $|y|\geq 0$

$\boxed{\text{y can be any real number.}}$

(45) $|b|\leq 0$

$\boxed{b=0}$

(46) $|x|<-1$

$\boxed{\text{No solutions}}$

The Absolute Value Inequality Theorems

In the following inequalities, C is a nonnegative constant.

I) If $|x|\geq C$, then

Algebraic Solution

$x\leq -C$ or $x\geq C$

Graphical Solution

II) If $|x|\leq C$, then

Algebraic Solution

$x\leq C$ and $x\geq -C$

also acceptable

$-C\leq x\leq C$

Graphical Solution

* If $<$ or $>$ signs appear, then open circles must be used (e.g. ——o——o—— or ——o——o——).

III) If $|x|<-C$, then there are no solutions.

IV) If $|x|\leq 0$, then $x=0$ only.

V) If $|x|\geq -C$, then x can be any real number.

VI) If $|x|>0$, then x can be any real number except zero.

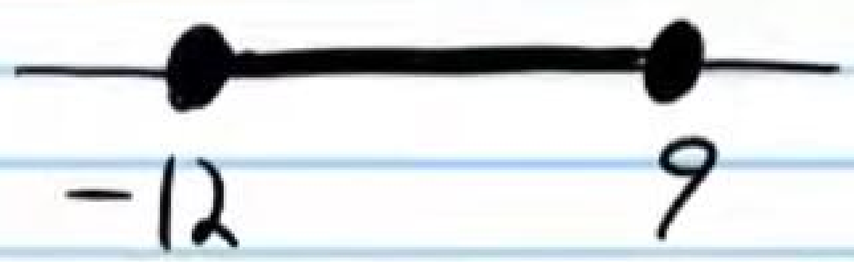
(47) $5 \ge \frac{|-2b-3|}{7} + 2$

$21 \ge |-2b-3|$

$-2b-3 \le 21$ and $-2b-3 \ge -21$

$-2b \le 24$ and $-2b \ge -18$

$b \ge -12$ and $b \le 9$

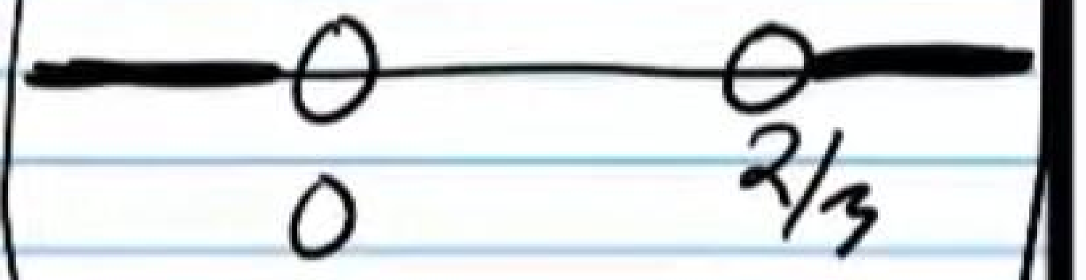


(48) $5|-2+6y|-8 > 2$

$|-2+6y| > 2$

$-2+6y > 2$ or $-2+6y < -2$

$y > \frac{2}{3}$ or $y < 0$



(49) $2\left|\frac{x}{-5}+6\right| + 3 \le 7$

$\left|\frac{x}{-5}+6\right| \le 2$

$\frac{x}{-5}+6 \ge -2$ and $\frac{x}{-5}+6 \le 2$

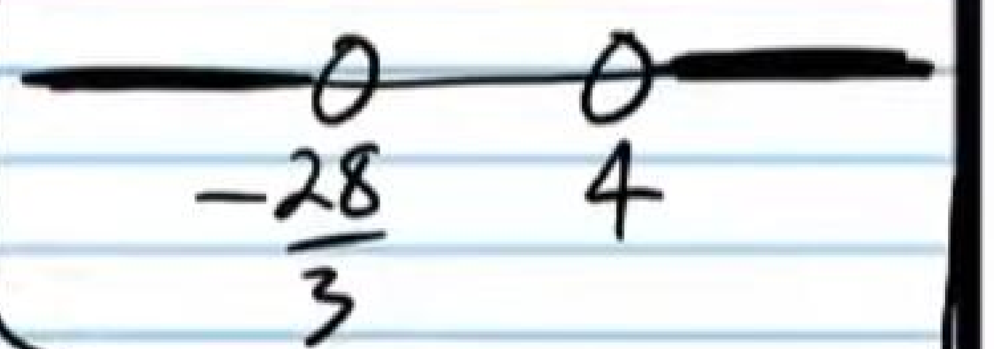
$x \le 40$ and $x \ge 20$

(50) $2 < -3 + \frac{|8+3y|}{4}$

$20 < |8+3y|$

$8+3y > 20$ or $8+3y < -20$

$y > 4$ or $y < \frac{-28}{3}$



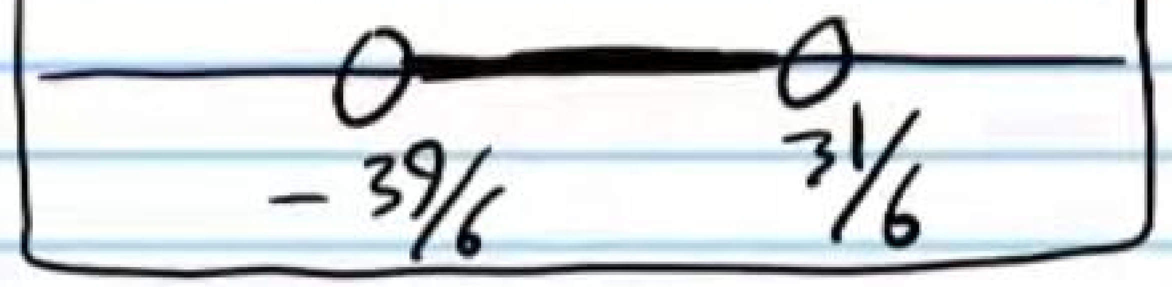
(51) $-5 < 2 + \frac{|4+6y|}{-5}$

$-7 < \frac{|4+6y|}{-5}$

$35 > |4+6y|$

$4+6y > -35$ and $4+6y < 35$

$y > \frac{-39}{6}$ and $y < \frac{31}{6}$



(52) $-4\left|\frac{b}{-6}-2\right| + 8 \le 7$

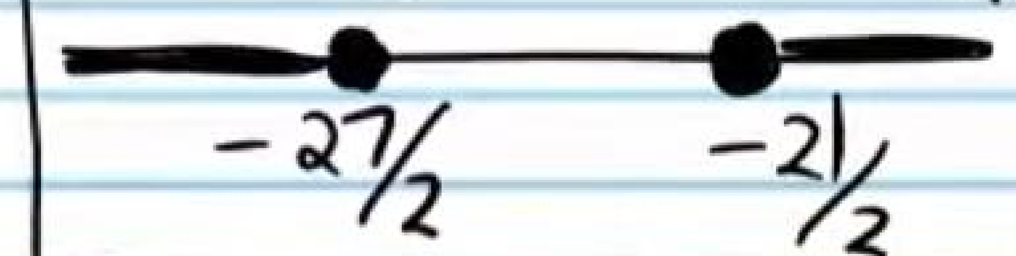
$-4\left|\frac{b}{-6}-2\right| \le -1$

$\left|\frac{b}{-6}-2\right| \ge \frac{1}{4}$

$\frac{b}{-6}-2 \ge \frac{1}{4}$ or $\frac{b}{-6}-2 \le -\frac{1}{4}$

$\frac{b}{-6} \ge \frac{9}{4}$ or $\frac{b}{-6} \le \frac{7}{4}$

$b \le \frac{-27}{2}$ or $b \ge \frac{-21}{2}$

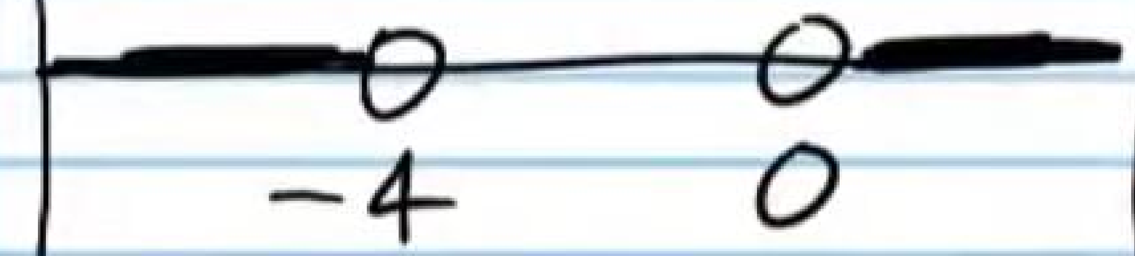


(53) $-44 > -8|6+3b| + 4$

$6 < |6+3b|$

$6+3b > 6$ or $6+3b < -6$

$b > 0$ or $b < -4$



(54) $\frac{3|2-5x|}{-4} + 6 < -1$

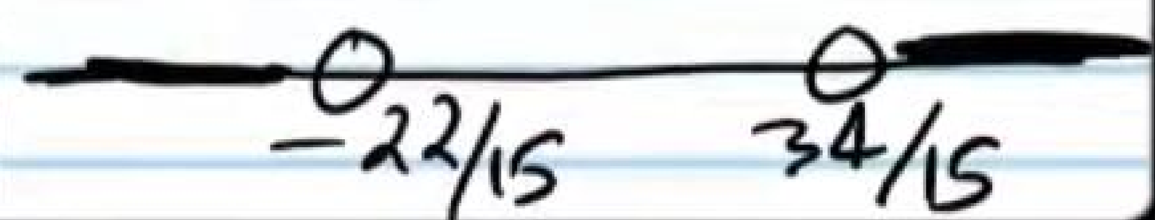
$3|2-5x| > 28$

$|2-5x| > \frac{28}{3}$

$2-5x > \frac{28}{3}$ or $2-5x < -\frac{28}{3}$

$-5x > \frac{22}{3}$ or $-5x < -\frac{34}{3}$

$x < -\frac{22}{15}$ or $x > \frac{34}{15}$



(55) $\dfrac{\left|\frac{b}{3}+4\right|}{-8}-10 \le -9$

$\left|\frac{b}{3}+4\right| \ge -8$

$\frac{b}{3}+4$ can be any real number, which means that b can be any real number.

(56) $8 \le 6-|8b-3|$

$2 \le -|8b-3|$

$2 \le (-1)|8b-3|$

$-2 \ge |8b-3|$

There is no number that 8b-3 can be that will work in the inequality, which means there is no value of b that will work either.

(57) $-2 \le -7\left|\frac{y}{-5}+1\right|-2$

$0 \le -7\left|\frac{y}{-5}+1\right|$

$0 \ge \left|\frac{y}{-5}+1\right|$

$\frac{y}{-5}+1=0$

$\boxed{y=5}$

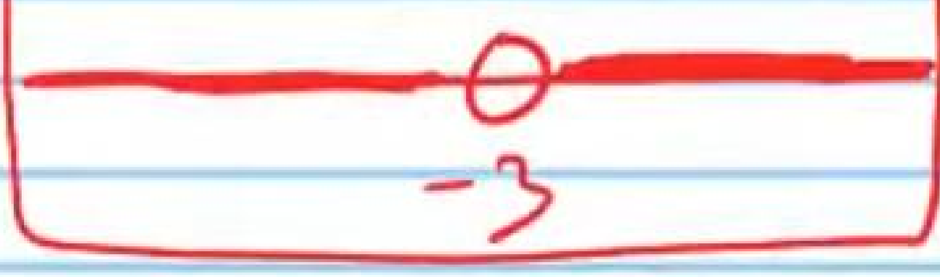
(58) $-1+\dfrac{|b+3|}{-2} < -1$

$\dfrac{|b+3|}{-2} < 0$

$|b+3| > 0$

$b+3 \ne 0$

$b \ne -3$



(59) $\dfrac{|5x+2|}{3}+4 < 4$

$|5x+2| < 0$

No solutions

(60) $-2|3-7y|+7 \ge 7$

$|3-7y| \le 0$

$3-7y=0$

$\boxed{y=\frac{3}{7}}$

(61) $-2|3x-1|+4 < 16$

$|3x-1| > -6$

X can be any real number.

(62) $8 < \dfrac{\left|\frac{y}{2}+5\right|}{4}+8$

$0 < \left|\frac{y}{2}+5\right|$

$\frac{y}{2}+5 \ne 0$

$y \ne -10$